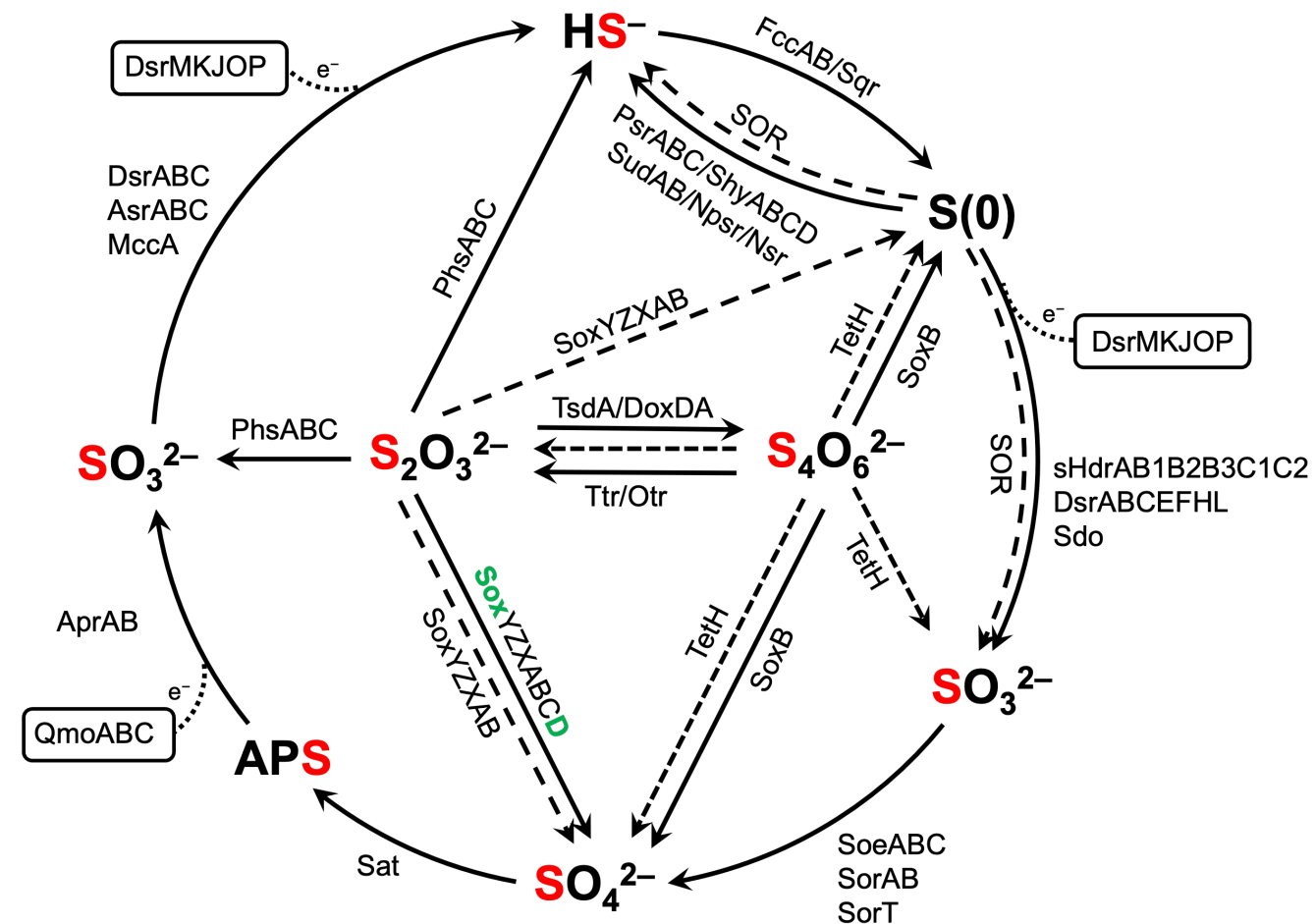


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The diagram illustrates the metabolic pathways of sulfur in a cell, divided into three compartments: Periplasm, Membrane, and Cytoplasm.

Periplasm:

- Top Left:** SoxB (blue oval) catalyzes the conversion of H_2O and SO_4^{2-} to 2H^+ and a disulfide-linked SoxYZ (orange box) complex, $-\text{O}_3\text{S-S-SoxYZ}$.
- Top Right:** SoxAX (blue oval) catalyzes the conversion of $\text{S}_2\text{O}_3^{2-}$ and a disulfide-linked SoxYZ complex to 2e^- and a mixed disulfide, SoxYZ-S-S-O_3^- .
- Bottom Left:** SoxCD (green oval) catalyzes the conversion of 6H^+ and a disulfide-linked SoxYZ complex to $3\text{H}_2\text{O}$ and 6e^- .
- Bottom Right:** SoxB (blue oval) catalyzes the conversion of 2H^+ and a mixed disulfide to SO_4^{2-} and a disulfide-linked SoxYZ complex.

Membrane:

- SoeABC (green box):** Catalyzes the conversion of $\text{MQ} \rightarrow \text{MQH}_2$.
- DoxDA (green box):** Catalyzes the conversion of $\text{MQ} \rightarrow \text{MQH}_2$.

Cytoplasm:

- TsdA (yellow box):** Catalyzes the conversion of $2\text{S}_2\text{O}_3^{2-}$ to $\text{S}_4\text{O}_6^{2-}$.
- SorAB/SorT (yellow boxes):** Catalyze the conversion of SO_3^{2-} to SO_4^{2-} .

Inter-compartmental Transport:

- A disulfide-linked SoxYZ complex is transported (indicated by a question mark) to the cytoplasm, where it is converted to R-S-S- .
- Dsr (blue oval):** Catalyzes the conversion of R-S-S- to HSO_3^- and SO_4^{2-} .
- sHdr (blue oval):** Catalyzes the conversion of HSO_3^- to SO_4^{2-} .

Sulfur Storage:

- type I:** sHdrB1, sHdrB2, sHdrC1, sHdrC2, sHdrA, sHdrA, sHdrH, sHdrI.
- type II:** sHdrB1, sHdrB3, sHdrC1, sHdrA, sHdrA, sHdrH.